A more complete and realistic description of the fall of gas streams toward a black hole was devised in 1969 by Donald Lynden-Bell, a British astrophysicist in Cambridge. Lynden-Bell argued, convincingly, that after the gas streams collide, they will join together, and then centrifugal forces will make them spiral around and around the hole many times before falling in; and as they spiral, they will form a disk-shaped object, much like the rings around the planet Saturn—san *accretion disk* Lynden-Bell called it, since the gas is "accreting" onto the hole. (The right half of Figure 8.7 shows an artist's conception of such an accretion disk around the modest-sized hole in Cygnus X-l.) In the accretion disk, adjacent gas streams will rub against each other, and intense friction from that rubbing will heat the disk to high temperatures.

In the 1980s, astrophysicists realized that the brilliant light-emitting object at the center of 3C273, the object 1 light-month or less in size, was probably Lynden-Bell's friction-heated accretion disk.

We normally think of friction as a poor source of heat. Recall the unfortunate Boy Scout who tries to start a fire by rubbing two sticks together! However, the Boy Scout is limited by his meager muscle power, while an accretion disk's friction feeds off gravitational energy. Since the gravitational energy is enormous, far larger than nuclear energy, the friction is easily up to the task of heating the disk and making it shine 100 times more brightly than the most luminous galaxies.

How can a black hole act as a gyroscope? James Bardeen and Jacobus Petterson of Yale University realized the answer in 1975: If the black hole spins rapidly, then it behaves precisely like a gyroscope. Its spin direction remains always firmly fixed and unchanging, and the swirl of space near the hole created by the spin (Figure 7.7) remains always firmly oriented in the same direction. Bardeen and Petterson showed by a mathematical calculation that this near-hole swirl of space must grab the inner part of the accretion disk and hold it firmly in the hole's equatorial plane—and must do so no matter how the disk is oriented far from the hole (Figure 9.6). As new gas from interstellar space is captured into the distant part of the disk, it may change the distant disk's orientation, but it can never change the disk's orientation near the hole. The hole's gyroscopic action prevents it. Near the hole the disk remains always in the hole's equatorial plane.